Using the ARMADA Algorithm to Analyze Physiological Marker Patterns in Presumed Emotional States

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Abstract

Affective computing has emerged as a pivotal field in bridging the gap between human emotional intelligence and machine responsiveness, focusing on the computational modeling and automatic recognition of human emotions. This study aims to investigate the feasibility of detecting sequential patterns of physiological transitions during presumed emotional episodes. To achieve this, the ARMADA algorithm is utilized to analyze datasets characterized by continuous annotations and synchronized cardiovascular and electrodermal features. Our findings include [to be completed based on experimental results]. [Final concluding sentence regarding the impact of the findings to be added here].

Keywords: Physiological Rule Mining, Emotion Recognition, ARMADA Algorithm, Affective Computing, Cross-Dataset Analysis

1. Introduction

Automatic emotion recognition plays a crucial role in Affective Computing, offering the potential for continuous assessment of human emotional states [25]. Emotion recognition systems are built on diverse modalities, most notably facial expressions and vocalisation [**?**]. Still, researchers have long also recognized the potential of physiological signals for emotion assessment. This stems from the fact that unlike facial expressions or voice, physiological responses such as Electrodermal Activity (EDA), Photoplethysmography (PPG), and Heart Rate Variability (HRV) are largely controlled by the autonomic nervous system and reflect genuine emotional states without conscious distortion [27]. In recent years, an extensive body of literature has adopted machine learning and deep learning techniques, including Support Vector Machines (SVMs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks such as LSTMs, to process these complex time-series signals [2]. Although these models often achieve strong classification performance, their predictions are typically difficult to interpret in mechanistic terms, making it challenging to explicitly explain how temporally evolving physiological changes relate to the emergence of a particular emotional state.

From another perspective, the past few years have highlighted the necessity of shifting emotion recognition systems away from 'black box' models toward architectures that enable explainability of their outputs [8, 1]. Despite the abundance of modern Explainable AI (XAI) techniques, rule-based systems have seen a resurgence in importance due to their naturally transparent and self-explanatory nature [5, 20]. The application of sequential rule mining to physiological signals, all the while, remains in its infancy. Even though human physiological responses naturally unfold over time, very few studies actually use sequential rules to analyze them [24]. To address this gap the research objective is to **explore the feasibility of discovering temporal patterns within specific physiological signals, leveraging continuously annotated data through the application of the Association Rule Mining And Discovery Algorithm (ARMADA)** [10, 30].

To achieve the goal, firstly, the research questions were defined in Section 2. Subsequently, a literature review is conducted in Section 3. Section 4 details the research methodology, while the subsequent sections follow this defined methodology directly, addressing the data acquisition

(Section 5), experimental design (Section 7) and implementation (Section 8), as well as the analysis of the results (Section 9). The final Section 10 concludes the study by summarizing the key findings and outlining directions for future work.

2. Research Questions

To provide a structured investigation to achieve the primary objective of this study, the research focuses on the following two research questions:

1. **RQ1: Technical Feasibility and Rule Stability**

   To what extent can the ARMADA algorithm effectively extract stable and statistically significant temporal association rules from multi-modal, continuously annotated physiological datasets?

   To address the complexity of this question, the following sub-questions are investigated:

   - **RQ 1.1 (Annotation Source)**
     How does the source of emotional labels—specifically self-reports versus expert external annotations—influence the quality and consistency of the discovered temporal rules?
   - **RQ 1.2 (Emotional Model)**
     To what extent does the choice of emotional model (discrete categories [6] vs. multidimensional valence-arousal space [22]) affect the algorithm's ability to identify meaningful patterns in physiological signals?

2. **RQ2: Pattern Discovery and Physiological Correlation**

   Which specific temporal patterns in physiological signals (such as HRV and EDA) are most frequently associated with transitions in emotional states according to the discovered rules?

   To analyze the generalizability of these patterns, the following sub-questions are investigated:

   - **RQ 2.1 (Demographic Factors)**
     To what extent do demographic characteristics, specifically gender and age, influence the structure and types of temporal association rules extracted by the ARMADA algorithm?
   - **RQ 2.2 (Cross-group Consistency)**
     Are there identifiable "universal" physiological patterns that remain consistent across different demographic groups, or are the discovered rules predominantly group-specific?
   - **RQ 2.3 (Multimodality Advantage)**
     How does the performance and interpretability of multi-modal association rules compare to single-modal rules in describing emotional transitions?

3. Related Work

3.1. Sequential Rule Mining in Affective Computing

While sequential pattern mining has been successfully applied to discover temporal relationships within data, its utilization in emotion recognition has been mostly restricted to other modalities. A significant amount of research has focused on extracting sequential emotion-aware rules from

text data, particularly social media content and micro-blogs [28, 29]. For instance, tracing the temporal emotional roadmap of users through sequences of online posts has proven valuable in depressive tendency detection [7]. Similarly, sequential pattern mining has been utilized to analyze multimodal combinations of verbal and facial clues in video data, unearthing temporal patterns during interpersonal interactions [19]. Despite these advancements across text and video modalities, physiological data remains notably excluded from sequential rule-mining approaches, leaving an unaddressed gap regarding the explicit, universally interpretable temporal rules defining human physiological reactions.

4. Methodology

This section outlines the research framework, starting with the fundamental assumptions that guide the analysis of physiological responses to emotional stimuli as well as core methodology depicted in Figure 1.

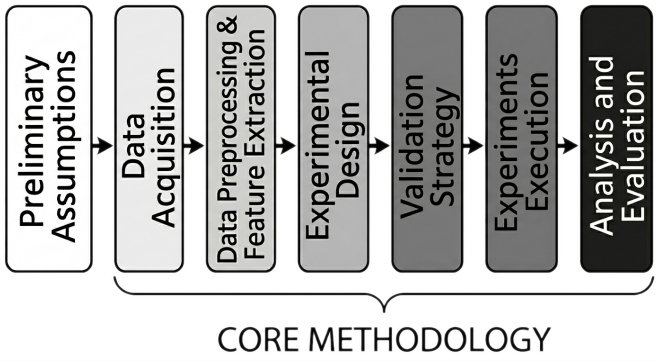

Fig. 1. Research methodology applied in the study.

4.1. Preliminary Assumptions

To ensure the clarity and focus of the investigation, the following primary assumptions and technical choices have been established:

1. **Selection of Physiological Modalities:** The study focuses on two primary types of signals: Cardiac Activity (specifically Heart Rate Variability – HRV) and Electrodermal Activity (EDA). These modalities were selected for two main reasons. First, they are highly sensitive to autonomic nervous system changes, providing reliable markers for emotional arousal and valence. Second, they represent the most prevalent biosignals in publicly available affective computing datasets, ensuring that the proposed framework is compatible with a wide range of existing research benchmarks and allows for cross-dataset validation of the discovered rules.

2. **Pattern Discovery Paradigm:** The research framework is based on the Sequential Association Rules (SAR) mining approach. In contrast to static association rules, SAR accounts for the temporal order of events, which is essential for capturing the dynamic nature of emotional transitions and the inherent physiological lag between a stimulus and the body's response.

3. **Algorithmic Framework:** The pattern discovery process is executed using the ARMADA algorithm, which is specifically engineered to extract association rules from multimodal time-series data. A fundamental characteristic of this approach is its reliance on

discretized symbolic intervals [30] rather than raw continuous data points. This methodology enables the seamless integration of signals with heterogeneous characteristics, such as high-frequency cardiac data and slow-varying electrodermal activity, by transforming them into a unified stream of interval-based events. By representing physiological changes as time-stamped symbols with a defined duration, ARMADA facilitates the identification of complex temporal predicates—such as *before*, *meets*, *overlaps*, *is-finished-by*, *contains*, *equals*, and *starts* — that extend beyond simple co-occurrence. This capability is essential for modeling the physiological lag inherent between an emotional stimulus and the subsequent autonomic nervous system response. Furthermore, the algorithm does not impose a rigid, a-priori structure on the discovered rules, allowing for the detection of both acute signal spikes and long-term correlational trends. The resulting set of human-readable "If-Then" rules ensures total model transparency, directly aligning with the XAI paradigm by providing causal-like justifications for affective state classifications.

4. **Rule Evaluation Metrics:** To assess the quality and reliability of the discovered patterns, the research framework relies on standard association rule metrics—primarily *Support*, *Confidence*, and *Lift*—specifically adapted for temporal sequences.

4.2. Core Methodology

4.3. Preliminary Assumptions

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Pattern Discovery Paradigm: We adopt the Sequential Association Rules (SAR) mining approach. Unlike static association rules, SAR accounts for the temporal order of events, which is critical for capturing the dynamic nature of emotional transitions and physiological lag.

Algorithmic Framework: The ARMADA algorithm is selected as the core mining tool. ARMADA is specifically designed for multi-modal datasets, allowing for the efficient discovery of complex, time-stamped relationships across heterogeneous signals without the need for a "black-box" architecture.

Selection of the ARMADA Algorithm

Selecting an appropriate data mining algorithm for this study required matching the specific nature of our dataset with the algorithm's technical capabilities. Specifically, the chosen method had to support multiple data channels simultaneously, integrate both categorical variables (e.g.,

participant sex) and continuous numerical values (e.g., EDA signals), and, most importantly, capture strict time consequences between physiological events and emotional responses.

Although some temporal rule-mining approaches can operate without an explicit discretization step [32], this is not problematic in our setting, as ARMADA is designed for discretized interval-based events [30]. This representation is also consistent with the nature of the analyzed features, which are subject-specific and normalized before rule extraction, making their transformation into discrete states both natural and interpretable.

Moreover, ARMADA is able to represent several types of temporal relationships between interval-based events, including *before*, *meets*, *overlaps*, *is-finished-by*, *contains*, *equals*, and *starts* [30]. Thus, the relation between two events is not limited to a simple ordering such as A before B, but can also capture more specific temporal configurations between intervals. This is useful in physiological emotion recognition, where the association between a physiological response and an emotional state may depend not only on event order, but also on the exact temporal relation between them. In addition, the `MAXGAP` parameter makes it possible to constrain the maximum allowable delay between related events, focusing the rule discovery process on temporally local dependencies [30].

5. Data Acquisition

Although a broader range of affective datasets exists [16], in this work we focused on those that combine multimodal physiological recordings with relatively continuous or densely sampled emotion annotations and for which data access could be obtained.

The analysis was conducted on five multimodal emotion datasets: EMBOA, K-emoCon, CASE, CEAP-360VR, and EmoWorker [12, 17, 25, 31, 14]. These datasets differ in both acquisition setting and annotation protocol, covering controlled laboratory recordings as well as conversational and immersive VR scenarios. They also include both self-reported annotations and externally assigned labels, which makes them suitable for evaluating rule mining under heterogeneous emotion-labeling schemes.

More specifically, EMBOA was collected in a human-robot interaction setting and combines Empatica E4 peripheral signals with externally assigned behavioral annotations [12], whereas K-emoCon captures dyadic conversations and provides self, partner, and external affective ratings alongside physiological recordings [17]. CASE represents a controlled laboratory dataset with high-resolution physiological signals and continuous self-assessment gathered using a joystick interface [25], while CEAP-360VR extends this paradigm to immersive 360° audiovisual stimulation in virtual reality [31]. EmoWorker, in turn, provides multimodal physiological recordings acquired in a workplace-related context together with continuous affect annotations [14]. The annotation schemes used in the evaluated datasets are summarized in Table 1.

Table 1. Annotation methodologies and frequencies across evaluated datasets.

Dataset	Annotation Type	Annotation Freq.
EMBOA	External (BORIS)	Variable
K-emoCon	Self, Partner, External	0.2 Hz (every 5s)
CASE	Self (Continuous Joystick)	20 Hz
CEAP-360VR	Self (Continuous + SAM Questionnaire)	10 Hz
EmoWorker	Self (Continuous traces)	5-10 Hz

5.1. Modalities

The datasets further vary in the physiological modalities and sampling rates they provide. While several of them share common peripheral signals, such as Electrodermal Activity (EDA) and

Blood Volume Pulse (BVP), others additionally include modalities such as ECG, EEG, respiration, eye tracking, or motion-related signals [12, 17, 25, 31, 14]. The physiological modalities considered in our experiments are listed in Table 2.

Table 2. Primary physiological modalities and sampling frequencies across evaluated datasets.

Dataset	Physiological Frequencies
EMBOA	EDA (4Hz), BVP (64Hz), ACC (32Hz), SKT (4Hz)
K-emoCon	EDA (4Hz), BVP (64Hz), ACC (32Hz), SKT (4Hz), EEG, ECG
CASE	EDA, BVP, ECG, RSP, EMG, SKT (All 1000Hz)
CEAP-360VR	EDA (4Hz), BVP (64Hz), ACC (64Hz), SKT (4Hz), Eye tracking
EmoWorker	EDA (4Hz), BVP (64Hz), ACC (32Hz), SKT (4Hz), ECG (130Hz), EEG (256Hz)

To facilitate a unified comparative analysis and enable the ARMADA-based rule mining process, data from all disparate sources was resampled, synchronized, and aligned into standardized 5-second multi-resolution time windows.

The common modalities used in the cross-dataset analysis were EDA, BVP, and skin temperature (SKT). These signals were selected because they are available in most of the analyzed datasets and provide some information about autonomic aspects of emotional responses.

Electrodermal activity (EDA) reflects changes in skin conductance driven mainly by sympathetic nervous system activation [11]. For this reason, EDA is one of the physiological modalities most consistently associated with arousal [11, 26].

Blood Volume Pulse (BVP) is a pulse-related physiological signal derived from photoplethysmography [15], from which features such as heart rate (HR), inter-beat interval (IBI), and heart rate variability (HRV) can be estimated. HRV features include, for example, SDNN, RMSSD, and pNN50 [13]. In the literature, BVP-derived features are more consistently associated with arousal than with valence [18, 11, 27].

Skin temperature (SKT) is a peripheral physiological measure influenced by blood flow in the skin and autonomic nervous system activity. In emotion research, it is treated mainly as an indirect marker of autonomic arousal [23].

6. Preprocessing

6.1. Sliding Window Mechanism

To synchronize disparate sampling rates across multiple datasets, continuous physiological signals and ground truth annotations were segmented using a static sliding window mechanism. We applied a multi-resolution methodology tailored to modality reactivity parameters: dynamic signals such as electrodermal activity (EDA), heart rate, and accelerometer data were extracted utilizing short 5-second sliding windows [3]. Conversely, parameters inherently reliant on longer macroscopic measurement phases, specifically heart rate variability (HRV) and skin temperature, were captured via independent 30-second windows [21].

6.2. Discretization of Physiological Features

Because ARMADA requires symbolic input, continuous physiological signals were discretized into three states: Low (L), Medium (M), and High (H). To reduce subject-specific bias, since physiological baselines and response ranges can differ markedly across individuals [9, 4], we first applied participant-wise min-max normalization and then defined L/M/H using global terciles computed from the pooled normalized data.

6.3. Normalization and Discretization of Emotional Labels

For datasets featuring native dimensional annotations (e.g., 1–9 SAM scales), arousal and valence values were normalized to a $[0, 1]$ range per dataset and divided into strict terciles: low ($[0, 0.33)$), medium ($[0.33, 0.66)$), and high ($[0.66, 1.0]$).

Conversely, the EMBOA dataset uses a discrete emotional model, with external observers attributing percentage agreements to six primary emotions (Happy, Sad, Scared, Disgusted, Surprised, Angry) via the BORIS software. To unify the data representation, these discrete labels were mapped to continuous dimensional coordinates using Russell's circumplex model (e.g., Happy: $V = +1, A = +0.5$; Sad: $V = -1, A = -1$). The weighted aggregates were then rescaled to $[0, 1]$ and similarly discretized. This translation successfully yielded 1489 valid valence states (929 *medium*, 518 *high*, 42 *low*) and 952 unified arousal states (711 *medium*, 229 *high*, 12 *low*) suitable for ARMADA processing.

To enable dimensional vs. discrete comparisons (Section 9), all dimensional data were mapped into a 3×3 grid of 9 categorical emotions via Russell's circumplex model. Combinations of arousal and valence terciles defined unique discrete states (e.g., *high* arousal and valence yielded *excited*; *medium* states yielded *neutral*). This synchronized all datasets—both inherently dimensional and dimensional-mapped discrete cohorts (EMBOA)—into a single 9-label framework. Across all datasets, this produced 4073 *neutral* tuples, followed by *calm* (947), *content* (938), *alert* (603), *stressed* (603), *upset* (553), *excited* (484), *sad* (304), and *relaxed* (207).

6.4. Temporal Alignment of Ground Truth Annotations

To enable strict association rule mining, annotation streams recorded at varied frequencies or by multiple observers were aggregated into a singular ground truth label per temporal window.

For **self-reported datasets** (e.g., CASE, K-emoCon), high-frequency or asynchronous ratings within a physiological window were simply averaged (linear mean) to yield one synchronized dimensional label.

For **externally annotated datasets** featuring multiple observers, independent ratings were fused into a consensus sequence before window aggregation. For K-emoCon (External), we used the dataset's native *aggregated_external_annotations*, where up to five continuous rating streams were aggregated via majority voting (see Code Availability). In EMBOA, the native percentage agreement across multiple BORIS raters served as the proportional weight mapping categorical emotions onto Russell's dimensional space. This ensured deterministic emotion tracking across all cohorts.

6.5. Handling Missing Values

Corrupted intervals and missing values were inherently filtered or marked natively by the original dataset providers, allowing our methodology to bypass artificial interpolation entirely.

6.6. Age Cohort Unification

To support demographic pattern extraction (RQ2.1) without overly fragmenting the subset populations, participant ages from all 121 analyzed subjects were unified into a balanced binary threshold rather than multiple narrow buckets. Participants were strictly divided into *Young* (age ≤ 25; $N = 60$) versus *Older* (age > 25; $N = 61$) groupings.

7. Experimental Design

RQ1.

Eksperyment 1

Wejście: ze wszystkich datasetów, dla każdego datasetu per użytkownik, czynność multi-

variate time series (wszystkie features z C i EDA (1), emocja (jak opisana - v, arousal znormalizowane (2), Emocja pochodzi z adnotacji -). Opisać, że oddzielne szeregi czasowe dla self i external (jeżeli są)

1 cała ale w dyskretnym

1.1. 1. tylko dla self (va) 2. tylko dla external (va)

1.2 1. tylko dla self (dyskretny) tylko dla self (dyskretny)

W tym eksperymecie nie ustalamy dedykowanych progów dla metryk tylko schodzimy 30 procent czy dedykowany do wszystkich czy nie.

Eksperyment 2

Robimy to samo, ale dla każdego datasetu oddzielnie

Jak będziemy analizować. K-emonCon jest traktowany 2x Analiza uzyskanych reguł - wszystkie 2, wszystkie 3 (jeżeli nie będzie wszędzie 0) to próbujemy 4, ewentualnie 5

tylko dla av

Eksperyment 3 4 z self (wyciągamy reguły) 2 z eksternal (wyciągamy reguły)

av + dyskretny?

RQ2

Eksperyment 4

Wejście:

The overall experimental workflow consists of five primary stages designed to extract, process, and evaluate temporal rules relating physiological signals to emotional states. The detailed pipeline is illustrated below:

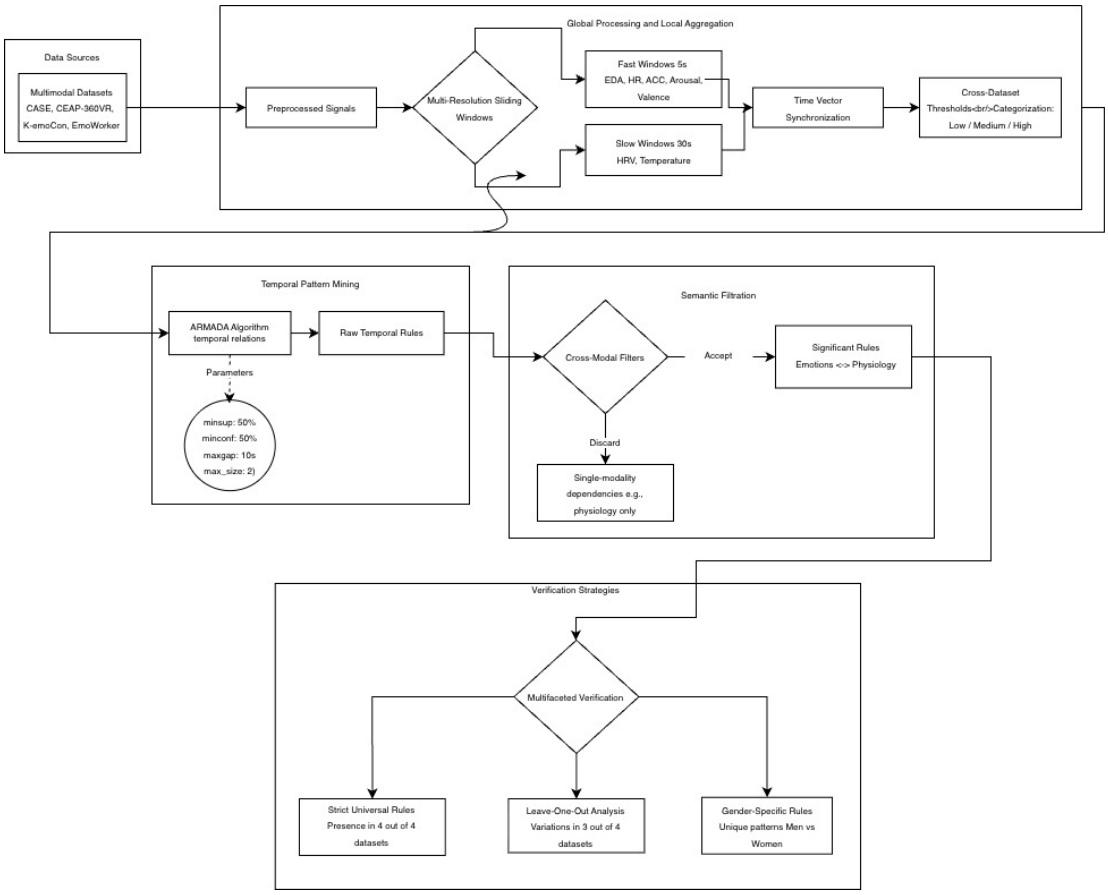

Fig. 2. The proposed experimental pipeline for extracting and verifying temporal physiological rules.

As outlined in Figure 2, the workflow involves the following steps:

1. **Data Sources:** The pipeline initially ingests multimodal physiological data from four diverse datasets: CASE, CEAP-360VR, K-emoCon, and EmoWorker.

2. **Global Processing and Local Aggregation:** The raw signals undergo global filtering, which includes peak detection and EDA decomposition, to derive clean preprocessed signals. These signals are then divided into multi-resolution sliding windows: 5-second fast windows for rapidly changing variables (EDA, HR, ACC) and 30-second slow windows for more stable measures (HRV, SKT). Finally, all time vectors are synchronized and the numerical data is discretized into cross-dataset thresholds (Low, Medium, High).

3. **Temporal Pattern Mining:** The categorized threshold data is fed into the ARMADA algorithm. Driven by specific support, confidence, and maximum gap parameters, the algorithm uncovers robust temporal relationships (such as *before*, *equals*, and *meets*).

4. **Semantic Filtration:** To ensure relevance to the study's objective, raw temporal rules are processed via cross-modal filters. Any rules that identify dependencies exclusively within a single modality (for example, a physiological change preceding solely another physiological change) are discarded, preserving only those significant rules that securely link an emotional state with corresponding physiological markers.

5. **Verification Strategies:** The remaining set of significant rules undergoes a three-pronged multifaceted verification phase: Strict Universal Rules analysis to find patterns holding across all four datasets, Leave-One-Out Analysis to spot major cross-dataset variations, and Gender-Specific checks to uncover unique behavioral variants between men and women.

8. Execution of Experiments

The ARMADA algorithm was executed independently on six physiological datasets to address the research questions defined in Section 2. Four datasets employ *self-annotations* (participants rate their own emotions): CASE ($N = 30$), K-emoCon ($N = 28$), CEAP-360VR ($N = 32$), and EmoWorker_v2 ($N = 31$). Two datasets employ *external annotations* (observers rate participants' emotions): K-emoCon external ($N = 28$, same physiological data as K-emoCon self) and EMBOA ($N = 16$, BORIS-based categorical annotations mapped to arousal/valence via Russell's circumplex model). All experiments were implemented in Python and executed on a standard workstation.

8.1. RQ1: Technical Feasibility

To assess the technical feasibility of temporal rule extraction from physiological data, the ARMADA algorithm was applied to all six datasets under identical strict parameters: $minsup = 0.5$, $minconf = 0.5$, $maxgap = 5$ s, and $max_pattern_size = 2$. These settings retain only rules that appear in at least half of the participants with at least 50% confidence.

In all experiments, the following semantic filters were applied to ensure cross-modal relevance:

- Rules containing only BVP/HRV/HR-related states were rejected.
- Rules containing only EDA-related states were rejected.
- Rules linking exclusively physiological modalities without any emotional dimension (arousal or valence) were rejected.
- Rules involving only a single feature type were rejected.

8.2. RQ 1.1: Annotation Source Comparison

To investigate the influence of annotation source on rule quality (RQ 1.1), the rules extracted from all four self-annotated datasets were pooled (union) and compared against the pooled rules from both externally annotated datasets. Additionally, a controlled comparison was performed on K-emoCon, where the same physiological recordings are available with both annotation types, isolating the effect of annotation perspective from physiological differences.

8.3. RQ 1.2: Emotional Model Comparison

To assess the impact of the emotional model on pattern discovery, all six datasets were processed under two distinct label representations with the same strict parameters ($minsup = 0.5$, $minconf = 0.5$, $maxgap = 5$ s). In the *dimensional* variant, arousal and valence were discretized into three levels (low, medium, high) and used directly as ARMADA events. In the *discrete* variant, arousal and valence were jointly mapped onto a 3×3 grid to produce nine categorical emotion labels based on Russell's circumplex model [22]: *stressed*, *alert*, *excited*, *upset*, *neutral*, *content*, *sad*, *calm*, and *relaxed*. Each dataset's native arousal/valence scale was normalized to $[0, 1]$ before mapping.

9. Results of Experiments

9.1. RQ1: Technical Feasibility and Rule Stability

The ARMADA algorithm successfully extracted temporal association rules from all six datasets under strict parameters. Table 3 summarizes the per-dataset extraction results.

Table 3. ARMADA rule extraction results per dataset.

Dataset	Annotation	N	Raw rules	Filtered
CASE	Self	30	516	140
K-emoCon	Self	28	383	82
CEAP-360VR	Self	32	254	114
EmoWorker_v2	Self	31	412	108
K-emoCon	External	28	382	115
EMBOA	External	16	249	80

Across all datasets, the algorithm consistently produced between 80 and 140 semantically filtered cross-modal rules per dataset. The extracted rules link physiological features (HRV, EDA) to emotional dimensions (arousal, valence) through diverse temporal relations (*meets*, *before*, *starts*, *equals*), confirming the algorithm's capacity to capture temporal dynamics inherent in physiology–emotion interactions, even under conservative parameter settings and across heterogeneous experimental protocols.

Figure 3 visualizes the pairwise overlap between datasets. The highest overlap occurs between CASE and EmoWorker_v2 (49 shared rules) and between CASE and K-emoCon (ext) (45 shared rules), while EMBOA shows the least overlap with other datasets (3–16 shared rules).

9.2. RQ 1.1: Influence of Annotation Source

To assess the influence of annotation source, the union of all rules from the four self-annotated datasets (348 unique rules) was compared with the union from the two externally annotated datasets (192 unique rules). Table 4 summarizes the cross-paradigm comparison.

A Jaccard similarity of 0.213 (95 shared rules) reveals a modest overlap, suggesting a core set of physiology–emotion patterns is robust to the annotation paradigm.

In a controlled comparison on K-emoCon (identical physiological data), external annota-

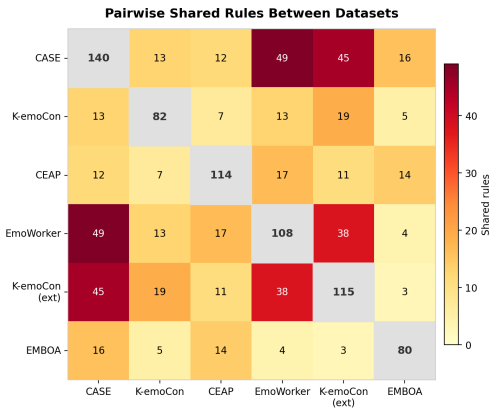

Fig. 3. Pairwise shared rules between all six datasets. Diagonal cells show the total number of filtered rules per dataset; off-diagonal cells show the number of rules shared between each pair.

Table 4. Cross-paradigm comparison: self-annotated vs. externally annotated rule sets.

Metric	Value
Total unique self rules (4 datasets)	348
Total unique external rules (2 datasets)	192
Shared rules	95
Self-only rules	253
External-only rules	97
Jaccard similarity	0.213

tions yielded 40% more rules (115 vs. 82, $J = 0.107$). Among the 19 shared rules, external annotations consistently showed higher confidence ($\Delta_{mean} = +0.102$). Table 5 lists the rules with the largest confidence shifts.

Table 5. K-emoCon shared rules with largest confidence differences between annotation sources.

Rule	Self Conf	Ext Conf	Δ
hrv_rmssd_high ⇒ meets valence_med	0.519	0.826	+0.308
hrv_cvnn_high ⇒ meets valence_med	0.556	0.857	+0.302
eda_medium ⇒ meets valence_med	0.577	0.875	+0.298
eda_scr_amp_high ⇒ starts valence_med	0.643	0.929	+0.286
arousal_low ⇒ meets eda_scr_auc_med	0.833	0.571	−0.262
arousal_low ⇒ is-finished-by eda_peaks_high	0.833	0.607	−0.226

These results suggest external annotations produce more consistent, higher-confidence rules, particularly for HRV→valence (confidence 0.83–0.93). Self-annotations, however, better captured certain arousal→EDA associations, hinting at higher sensitivity to interoceptive arousal. Nevertheless, the 95 rules shared across paradigms demonstrate that ARMADA extracts generalized patterns independent of annotation source.

9.3. RQ 1.2: Influence of Emotional Model

Table 6 compares the dimensional and discrete emotion models across all six datasets. Only cross-modal rules linking an emotional component to a physiological signal were retained.

At the standard support threshold (50%), the discrete model produced fewer cross-modal rules in every dataset (22–51% of the dimensional count). Neither model yielded any rules

Table 6. RQ 1.2: Dimensional vs. discrete emotion models per dataset ($minsup = 0.5$).

Dataset	Ann.	Dim Filt	Disc Filt	Ratio
CASE	Self	140	72	0.51
K-emoCon	Self	82	18	0.22
CEAP	Self	114	28	0.25
EmoWorker_v2	Self	108	35	0.32
K-emoCon	Ext	115	46	0.40
EMBOA	Ext	80	36	0.45

shared across all six datasets simultaneously. Pairwise Jaccard similarity was consistently higher for the dimensional model (mean $J = 0.090$) than for the discrete model (mean $J = 0.040$), indicating that the dimensional representation produces more consistent patterns across heterogeneous datasets. The discrete model's lower cross-dataset agreement reflects the inherent sparsity of a nine-category mapping, where peripheral emotions generate dataset-specific rather than universal rules.

9.4. RQ2: Pattern Discovery and Physiological Correlation

To identify temporal patterns mapping physiology to emotions consistently across diverse experimental conditions, a strict intersection analysis was performed on the four self-annotated datasets. At the standard 50% support threshold, no rules emerged in all four datasets simultaneously. Lowering the minimum support to 10% (while retaining 50% confidence) revealed exactly nine universal rules, all belonging to a single family: heart-rate variability metrics temporally coinciding with moderate emotional states.

Table 7 summarizes the discovered universal patterns.

Table 7. Universal physiological-emotional patterns (across 4 datasets at 10% support, 50% confidence).

Universal Pattern	Avg. Conf	Avg. Sup
hrv_hfn_high ⇒ meets valence_medium	0.654	0.451
hrv_rmssd_low ⇒ meets valence_medium	0.604	0.484
hrv_pnn50_high ⇒ meets arousal_medium	0.603	0.518
hrv_cvnn_high ⇒ meets arousal_medium	0.594	0.525
hrv_pnn50_low ⇒ meets arousal_medium	0.593	0.490
hrv_cvnn_medium ⇒ meets arousal_medium	0.591	0.500
hrv_pnn50_low ⇒ meets valence_medium	0.581	0.476
hrv_sdnn_high ⇒ meets arousal_medium	0.564	0.508
hrv_rmssd_low ⇒ meets arousal_medium	0.563	0.449

All nine universal rules share the temporal relation *meets* and exclusively associate HRV features with moderate arousal or valence states. Notably, no EDA-based rules achieved universality, suggesting that while electrodermal activity varies substantially across experimental paradigms, the HRV→emotion pathway remains the single most robust psychophysiological anchor across diverse datasets.

9.5. RQ 2.1 & 2.2: Demographic Influences and Cross-group Consistency

To evaluate the influence of demographic characteristics (RQ 2.1) and overall cross-group consistency (RQ 2.2), participants across the four self-annotated datasets were segmented by Gender and Age under a 10% minimum support threshold. Table 8 summarizes the volume of univer-

sally discovered rules within each demographic subgroup compared to their intersection.

Table 8. Cross-dataset rule counts specifically universally valid within demographic subgroups.

Split Category	Subgroup 1	Subgroup 2	Shared Overlap	Global Baseline
Gender	Male: 114	Female: 126	45	9
Age (≤ 25 **vs.** > 25)	Young: 122	Old: 109	36	9

While a foundational psychophysiological baseline exists (the 9 global rules), fine-grained temporal responses remain substantially demographic-specific. Evaluating data exclusively within demographic subgroups dramatically increases the number of consistent temporal rules (from 9 globally to 114–126 per gender group and 109–122 per age group). The intersection between subgroups (45 shared gender rules, 36 shared age rules) is considerably larger than the global baseline, indicating that homogeneous populations enable the discovery of more stable patterns. Nevertheless, the majority of subgroup-specific rules (69 unique to males, 81 unique to females) remain exclusive to their respective demographic, structurally demonstrating the influence of individual differences on physiological-emotional temporal dynamics.

10. Conclusions

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